

Technology News

Smart clothes to help you cope with stress

Smart clothes that monitor your psychological states, temperature and heart rate could soon be available so as to help you cope with stress.

As part of an artistic project called *Wearable Absence* uses garments, the clothes will be made of textiles woven with various wireless sensors that can track various types of biological conditions of the wearer, such as temperature, heart rate, breathing and galvanic skin response. The information received will be transferred to a Web-based database connected to the clothes. The data will then be analysed to work out the emotional state of the wearer. The database will then send media, such as songs, words and images, to the display and speakers in the clothes to calm the wearer. The simple display can handle small images and

scrolling text, while the speakers will be able to replay music, sounds or play recorded messages from friends or family.

The project involves two teams of researchers, led by Prof. Barbara Layne of Concordia University in Montreal, Canada, and Prof. Janis Jefferies of the University of London.

Collision-free cars

Ever wished you had a car that drove on its own while you just relaxed listening to music? Well, your wish might come true soon! Volvo has tested its new S60 model, which has the first in-car system that can sense an imminent collision with pedestrians and put brake automatically if the driver doesn't. The system is the latest in a line of developments made possible by sophisticated sensors based on cameras, radar and lasers.

The sensors already provide drivers with adaptive cruise control, which change the car's speed to maintain a safe distance from the vehicle in front, as well as technology such as semi-autonomous parking systems.

Jonas Ekmark, a researcher at Volvo headquarters near Gothenburg, Sweden, says that we are entering an era in which vehicles will also gather real-time information about the weather and highway hazards, and use this to improve fuel efficiency and make life less stressful for the driver and safer for all road users.

Bioprinter for human organs

In what could be good news for transplant patients, US scientists have developed a machine that can weave human organs with fragments of tissues. The researchers claim the doctors could one day be able to 'print' new organs for transplant. The method, based on 3-D laser printing technology, could be used to create body parts. Instead of combining layers of plastic and metal, the 'bioprinter' puts living tissues together.

Two laser-based printing heads are used to place living cells onto thin sheets of gel with microscopic precision. Multiple layers are then laid on top of each other in a specially designed mould, or 'scaffold', and the cells begin to fuse together.

"Ultimately, the idea would be for surgeons to have tissue on demand for various uses," said Keith Murphy, chief executive of the California-based company Organovo.

Graphene set to change future of electronics

Scientists who work with nano circuits are enthusiastic about finding a way to mass-produce graphene—a rival of silicon that will eventually revolutionise electronic devices ranging



Wearable Absence garments woven with sensors